Problem

- Takeaway meals generate large amounts of single-use food packaging waste
- Existing BYOC initiatives depend on students remembering to bring their own containers
    - Students who forget their containers have no alternative for reusable packaging

Solution

- Introduce a deposit-based reusable container sharing system across NTU canteens
    - Canteens stock durable and reusable containers and stainless-steel utensils
    - Participating canteens & number of containers available at each location will be integrated into NTU OmniBus app
        - Increases convenience in finding return locations that are nearby and not full
- How it works:

1. User pays a refundable deposit when borrowing a container from the machine
2. Pass the container to the stall owner to take away
3. Return it to any participating canteen at designated return spots as long as machines are not full (based on map)
4. Users can receive their deposit back anytime, as long as the borrowed container is returned
5. Canteen cleans using commercial dishwashers and redistributes to other canteens

- Benefits:
    - Complements BYOC initiatives by providing a convenient alternative for users who did not bring their own container
    - Reusable containers reduce climate change impact by 59% after 20 uses (Aggarwal, 2024)

Feasibility

Item	Container	Utensils	Washing	Total
Estimated Cost	$3	$1.50	$0.05/wash	$4.50

Estimated cost of disposable takeaway set: $0.13
Reusable set becomes cost-effectuive after approximately 56 users
Feasbility depends on high container return rates, low loss and damage rates, number of times containers are reused and actual labour costs

Limitations

- High initial cost due to upfront investment in reusable containers and utensils
- Container low return rate may result in higher replacement costs and eat into environmental benefits
- Additional washing results in more water and electricity used. May even incur higher manpower costs
- Containers need to be redistributed frequently
- Hygiene concerns whereby users may feel it is dirty or not cleaned properly

Implementation

- Conduct a pilot programme to test and modify the system based on user feedback
- KPIs to track:
    - Number of containers borrowed, returned, average number of times a container is reused, number of damaged/lost containers, actual washing & replacement costs, reduction in takeaway containers used
- Surveys to be conducted on canteen users, stall owners and staff at the end of the programme to identify user concerns and areas for improvement

Expected outcomes

- Reduction in single-use takeaway packaging
- Increase accessibility of reusable packaging
- Encourage sustainable habits through regular borrowing and returning
- Lower environmental impact and carbon footprint from canteens